Sara Razavi

sararazavi154@gmail.com • (225) 367-0858 • https://www.linkedin.com/in/sara-razavi • https://srazavi154.github.io/

EDUCATION

California Institute of Technology (Caltech) **Pasadena, CA**
Bachelor of Science, Major in Mechanical Engineering, Minor in Biology September 2021 – December 2025

Honors & Awards: Glacier Peak Summit Scholar; Caltech Bill Gross Entrepreneurship Award; Caltech Diversity & Equity Award; Caltech Engineering Competition Finalist; NASA Scholar; QuestBridge National Scholar; Johns Hopkins Global Health Leaders Conference Speaker

PROFESSIONAL EXPERIENCE

Cedars-Sinai **Los Angeles, CA**
Data Scientist March 2026 – Present

- Deployed production-grade machine learning architectures, integrating supervised and unsupervised learning frameworks, including XGBoost, Random Forest, and SVM, to automate clinical decision-making and optimize hospital operational workflows
- Constructed end-to-end, automated database-to-deployment pipelines within Oracle and Redshift environments by leveraging advanced Python and SQL to facilitate repeatable, high-performance longitudinal analyses and feature engineering across complex EMR datasets

Caltech-CERN **Pasadena, CA**
Precision Detector Systems Engineering Research Assistant October 2025 – January 2026

- Engineered and validated precision detector modules for CERN's CMS Barrel Timing Layer, integrating LYSO: Ce crystal arrays with SiPM sensors to achieve ultra-fast 30–60 ps timing resolution while optimizing end-to-end DAQ performance for next-generation particle tracking

Northrop Grumman **Dulles, VA**
Mechanical Systems Integration & Testing Engineer Intern June 2025 – August 2025

- Enhanced mechanical integration and testing workflows for Cygnus and MRV by synthesizing complex systems data into actionable insights, reducing integration issues by 20% and improving milestone alignment across cross-functional engineering, PM, and QA teams

Caltech – ME11C: Fluid Dynamics & ME 13/113: Mechanical Prototyping **Pasadena, CA**
Mechanical Engineering Teaching Assistant September 2023 – June 2025

- Mentored 60+ students in fluid dynamics, mechanical design, and prototyping, applying FEA principles and hands-on iteration to boost analytical problem-solving by 25%, enhance prototype performance by 15%, and cut failure rates by 10%

Cedars-Sinai **Los Angeles, CA**
Biomedical Device R&D Engineer Intern July 2024 – March 2025

- Architected automated data preprocessing pipelines for 20+ novel EIT-based electrode configurations, cutting research processing time by 40% and boosting ML model accuracy by 30% to advance clinical feasibility assessments

Caltech – Advanced Mechanical Bipedal Experimental Robotics Lab **Pasadena, CA**
Robotics Systems Research Engineer Assistant October 2023 – January 2024

- Customized and validated Myo Suite biomechanical simulations to generate patient-specific rehabilitation models, enhancing adaptability and scalability across neuromuscular profiles and advancing data-driven personalized therapy design

PROJECTS

Melano Metrics (MedTech Venture) **Los Angeles, CA**
Co-Founder & Product Design Lead March 2025 – June 2025

- Directed product design cross-functionally for an ML-powered solution correcting known SpO_2 reading inaccuracies in pulse oximeters and wearables for high-melanin individuals (Fitzpatrick V–VI), delivering a validated early-stage proof-of-concept
- Developed and pitched investor deck, articulating product vision, technical milestones, and FDA Breakthrough Devices Program as part of GTM, successfully securing $5K in competitive non-dilutive prize funding to support R&D

Caltech Bot Hockey Team **Pasadena, CA**
Lead Systems & Strategy Engineer September 2024 – March 2025

- Spearheaded the design and integration of high-performance mechanical subsystems, refining torque-optimized drivetrains and precision shooting mechanisms through iterative, data-driven testing, resulting in a finalist placement with three robust robots

SKILLS & INTERESTS

Skills: C++, Java, Python, JavaScript, HTML/CSS, SQL, CAD (SolidWorks, AutoCAD, Creo), 3D Printing, GD&T, Machine Shop, Ansys, ROS, MATLAB, Microsoft 365 Suite (Excel, PowerPoint, Word, SharePoint, Power Apps), Adobe Creative Cloud, UI/UX Design

Interests: Philosophy, Slam Poetry, Escape Rooms, Storytelling, Taekwondo (Black Belt), Oil Painting, Pastel, Horseback Riding, National Parks